



— PRE-SEED · 2026 · CONFIDENTIAL

Nuclear Propulsion as a Service

Decarbonizing global shipping with detachable offshore nuclear power.

DIMITRIS KOUTENTAKIS · EMILE GERMONPRÉ

THE PROBLEM

Ships burn the dirtiest fuel.

Bunker fuel is a thick, black sludge that must be heated just to flow.

1 GTn

GLOBAL CO₂ EMISSIONS

Shipping would be the 6th largest emitter if it were a country.

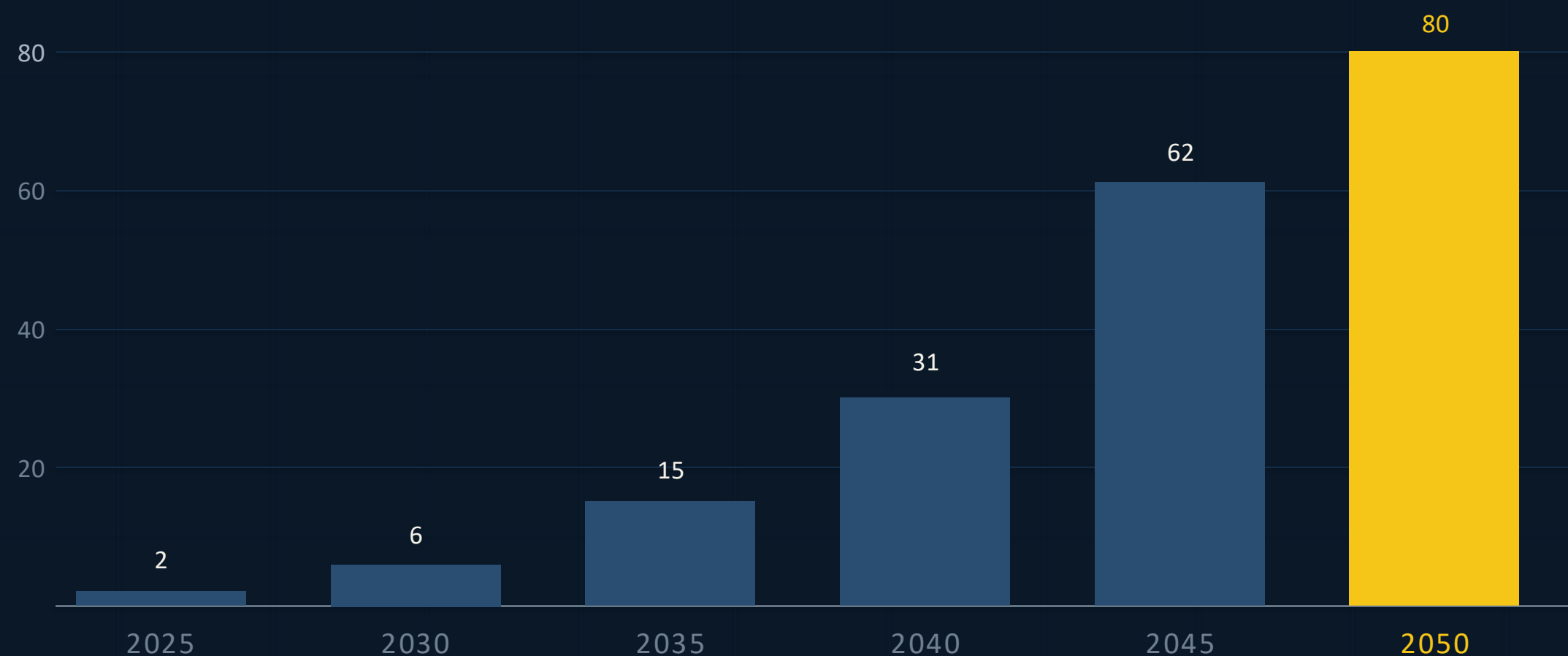
100K+

PREMATURE DEATHS / YEAR

From SO_x and NO_x pollution caused by burning HFO.

FUELEU MARITIME PENALTY INTENSITY

% REDUCTION REQUIRED VS 2020



THE PROBLEM

Bunker fuel's economics are getting worse

Slow-steaming is a temporary workaround

2-3x

FUEL PRICE CHANGES IN MONTHS

Volatility is the main OPEX risk.

>30%

SPEED REDUCTION

Vessels operating below design speed.

40x

FUELEU PENALTY RAMP BY 2050

Penalty intensity from baseline year.



SOURCES · CLARKSONS · IEA · IMO GHG STUDY 4 · FUELEU MARITIME REGULATION (EU 2023/1805)

Nuclear is the only viable alternative

NUCLEAR

The only proven option.

- ✓ Scalable supply chain
- ✓ Proven: 60+ years at sea
- ✓ Cost-effective at scale
- ✓ 100% zero-emission
- ✓ Maintains full speed

But conventional nuclear shipping has been blocked for 60 years.

X Alternative fuels (biofuel / hydrogen / ammonia / methanol)

Limited supply. Unproven at scale. 3–5× fuel cost.

X Wind / sails

Supplemental only. Cannot propel deep-sea vessels independently.

X Battery power

Weight is prohibitive for long-haul shipping. No port charging infrastructure.

X Carbon Capture

Unproven at maritime scale. Adds weight, cost, and complexity.

Non-technical barriers have blocked nuclear shipping for 60 years

01

Denied Access

Nuclear ships denied entry to commercial ports worldwide. Emergency zones around population centers make it impossible.

02

Slow regulation

Every state demands separate licensing. Onshore reactor licensing can take a decade.

03

High cost

Diseconomies of scale due to low power and low utilization and billion-dollar liability on shipowners balance sheet

04

Liability

Billions in nuclear liability per vessel. No shipowner will take this on their balance sheet.

Infeasible

Unattractive

Our LeadShip powers a convoy of vessels in the high seas

Customer ships keep their own engine and fuel

- 01 **Depart**
Ship leaves port under its own power.
- 02 **Rendezvous**
Meets LeadShip at a waypoint, beyond International Waters.
- 03 **Connect**
Dynamic ship-to-ship cable
- 04 **Transit**
99% of voyage under nuclear power · 15+ kn.
- 05 **Disconnect**
Before territorial waters; ship enters port alone.



LEADSHIP + CONVOY · DYNAMIC HV CABLE

COMPETITIVE EDGE

Our solutions makes nuclear shipping low-friction

BARRIER		ATOMARINE RESOLUTION	
X	Port access denied	→	✓ LeadShips never enter territorial waters; no constituency to oppose
X	Slow multi-national regulation	→	✓ Single design to license with a single Flag State
X	High cost	→	✓ Convoy economics drive costs down
X	Liability	→	✓ Zero nuclear liability on shipowner balance sheets.

Every barrier is an existential risk to our competitors. Our technology avoids them altogether.

Our customers get more than carbon-free energy: we provide stability, optionality, and lower costs

REVENUE MODEL

Atomarine delivers nuclear
propulsion as a service

PRICING Per-voyage fee indexed to energy + avoided carbon penalties.

CONTRACTS Long-term take-or-pay, secured by EU carbon-tax savings.



Lower fuel cost

Convoy economics undercut fossil fuel prices; the carbon upside is enormous



Cost stability

Multi-year fuel cycles result in cost stability



Fuel optionality

Hybrid ships can operate on other fuels (HFO/H₂/NH₃); no need to 'bet' on us



Easy adoption

Minimal newbuild cost addition (~\$2-5M) with PTI becoming newbuild standard
Zero nuclear liability and no nuclear expertise needed

Sharing the LeadShip cost across multiple vessels unlocks attractive economics

\$0.6-1.3B

LEADSHIP CAPEX NOAK / FOAK

40 yr reactor design life; NOAK shown for N = 15

14-21%

INTERNAL RATE OF RETURN

Without / with carbon taxes, 25 yr horizon

\$110-200M

ANNUAL VALUE CREATED PER LEADSHIP

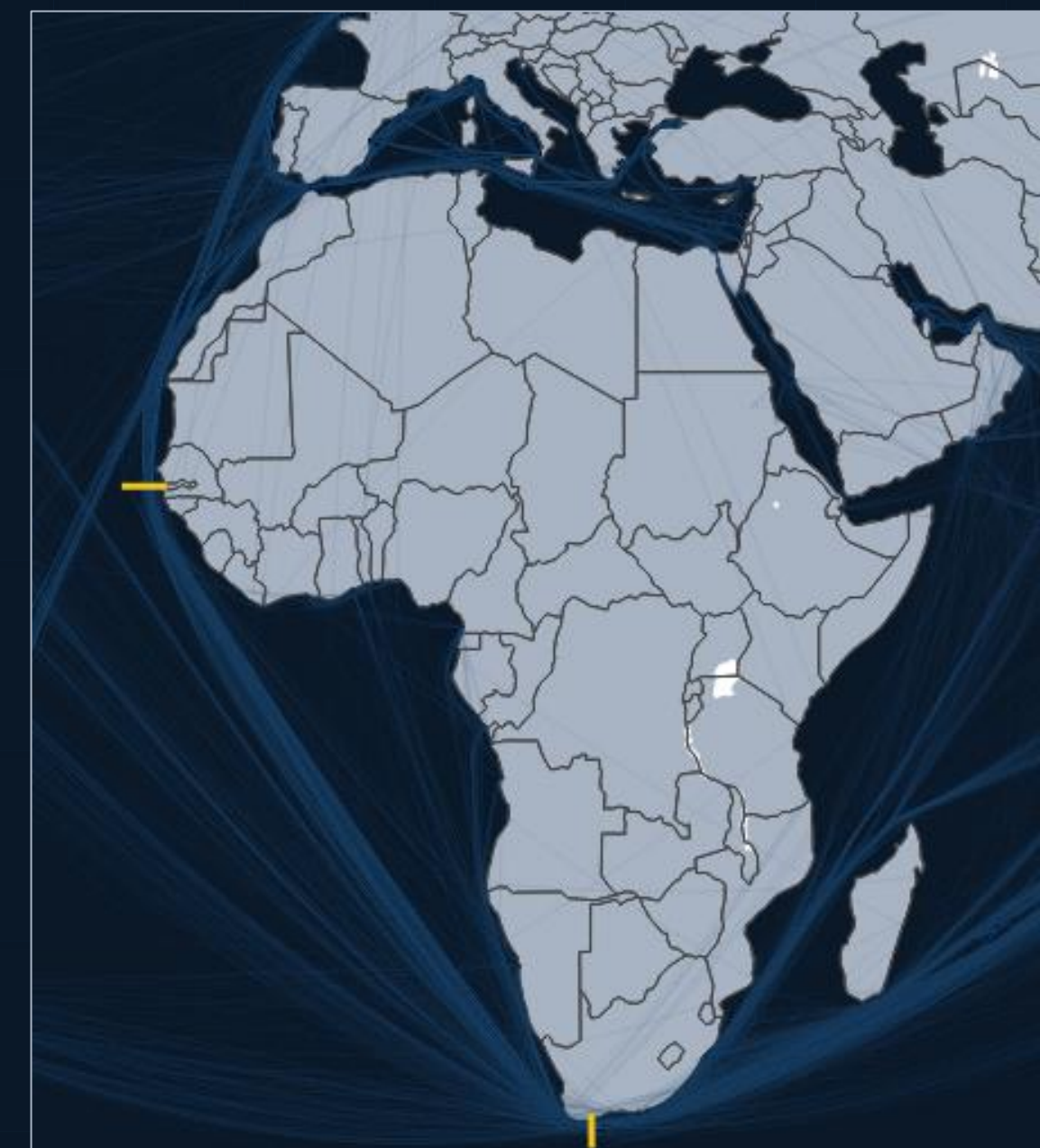
Minimum payback: 5-10 yr (full value capture, no customer discount)

\$450/ton

OPERATIONAL BREAKEVEN PRICE

\$640/ton for 20 yr payback, <\$0/ton with carbon taxes included

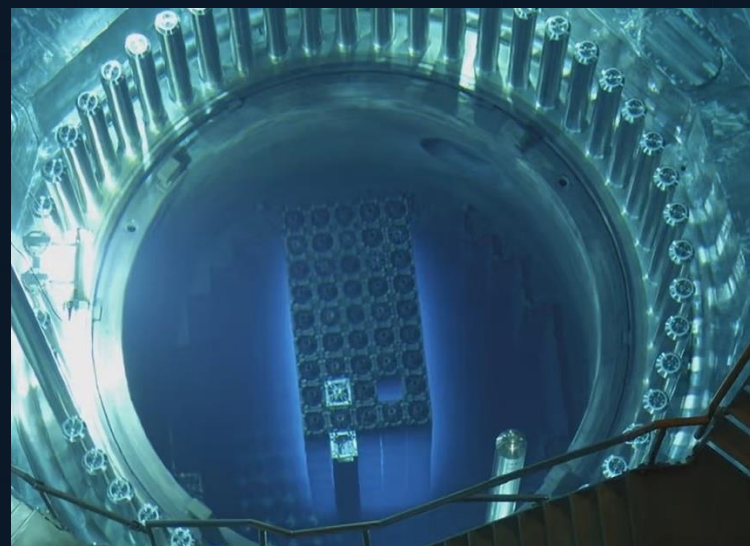
Modeled based on 2025 AIS data for >7k TEU ships · Conservatively, no rerouting assumed



REFERENCE WEST-AFRICA ROUNDING · 3 LEADSHIPS EN ROUTE

Recent technologies enable Atomarine

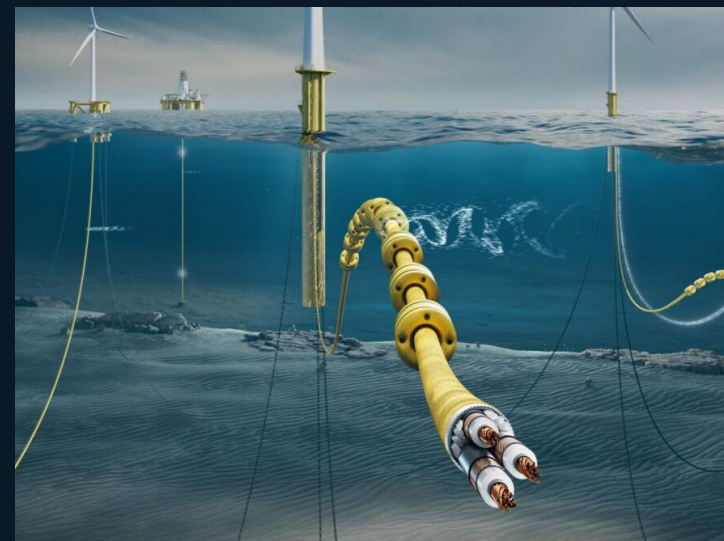
PILLAR 01



SMR Supercycle

MARINE IPWR REACTORS
160+ NAVAL VESSELS

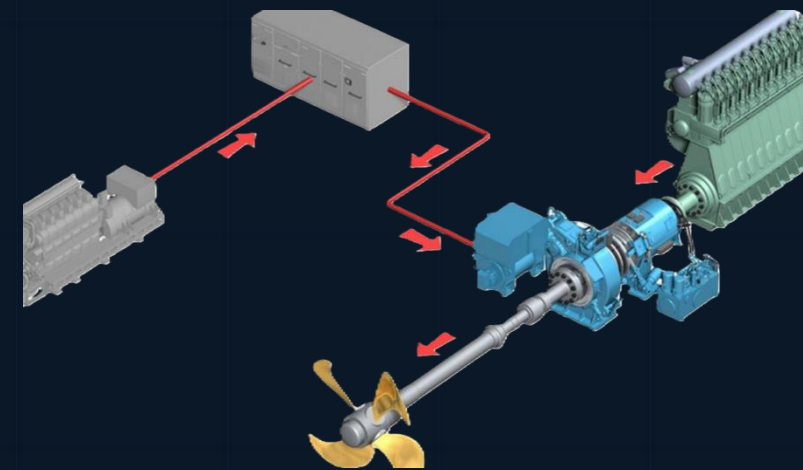
PILLAR 02



Dynamic Offshore Cables

PROVEN IN FLOATING WIND & OIL/GAS

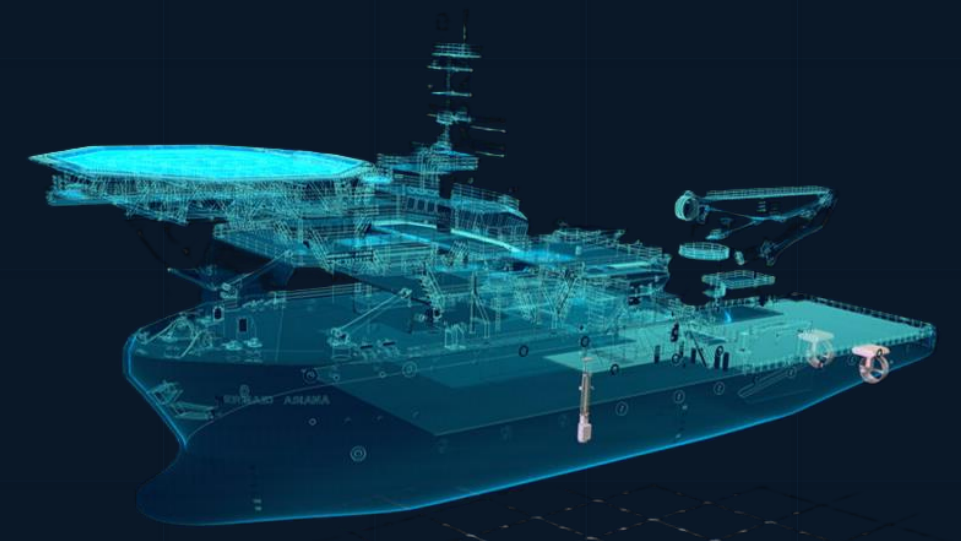
PILLAR 03



Electric-Drive Propulsion

NEWBUILD STANDARD
MODEST RETROFIT COST

PILLAR 04



Dynamic Positioning

METER-LEVEL ACCURACY

— VISION

We start by powering cargo ships and end with mobile nuclear infrastructure

HORIZON 1

TODAY → 2040

Container convoys

LeadShips decarbonize major deep-sea container routes where ETS exposure bites first.

HORIZON 2

2040 → 2050

Shipping shapes around us

Rerouting logistics to be convoy efficient. Ports adopt shore-side electric tugs;
Shipping is served by autonomous feeder barges that receive power from Atomarine LeadShips.

HORIZON 3

2050 +

Mobile nuclear utility

Offshore compute, naval logistics, disaster response, remote industrial loads: a new category of utility.

MARKET

A \$127B+ market under existential pressure.

TAM

\$130B + \$100B

ANNUAL SHIPPING FUEL SPEND

FUELEU CARBON COSTS

IMO carbon credits increase market size further

SAM

\$95B

BULK + TANKERS · ROUTES > 1,000 NM

75% of fuel consumption.

SOM

\$1.5B+

CONTAINER SHIPS · EU-REGULATED ROUTES

First wedge into the market.

BEYOND MARITIME

Floating datacenters

Centrally build self-powered datacenters and ship them to cool locations.

Remote energy supply

Plug in to any coastal grid; provide carbon-free energy to desalination plants.

Deep Sea Mining

Carbon-free power for hard-to-abate offshore loads.

Defense & logistics

Charge electric autonomous vessels; power deployments and emergency response.

At scale, Atomarine is irreplaceable

01

Network effect

More LeadShips → shorter wait times
→ better service.

02

Technology lock-in

Ships designed around our connectors.
High switching costs.

03

Intellectual property

Provisional patents filed. Multi-
jurisdiction filings planned.

04

Talent moat

First to assemble a nuclear-maritime
cross-disciplinary team.

Expertise and market access.



Dimitris Koutentakis

CO-FOUNDER & CEO

MIT MBA · Naval Engineering MS

MIT EECS Undergrad · Meng

Amazon electrification optimization

Greek shipping background



Emile Germonpré

CO-FOUNDER & CTO

MIT Nuclear Engineering PhD

Nuclear reactor systems specialist

Valedictorian · UGent Engineering

Fulbright Scholar

ADVISORS



Jacopo Buongiorno

MIT Professor Nuclear Engineering

Nuclear Lead MIT Maritime
Decarbonization Consortium

KEY AFFILIATIONS

MIT MTC

THE ENGINE

CLEANTECH OPEN

Progress made on key areas of risk

Building our moat and ecosystem

USPTO

PROVISIONAL PATENT

X-Energy, Apollo Atomics

LETTER OF SUPPORT

Validating customer traction

Costamare, Dorian LPG, MPCC

DATA-EXCHANGE

LOIs

TARGETED NEXT 6 MONTHS

Proving the business case and logistics

TEA v9

BREAKEVEN AT N=2

Convoy logistics

GLOBAL POSITIONS MODEL

Fast-tracking regulatory pathway

ABS, BV, DNV

CLASSIFICATION ENGAGEMENT

NRC

EARLY DISCUSSIONS

Showing convoy feasibility

Cable dynamics

EXPERT VALIDATION

Autonomous station-keeping

EXPERT VALIDATION

Staged de-risking to first commercial LeadShip.



— THE WINDOW IS NOW

The next century of global shipping is nuclear.

We replace the fuel layer of shipping.

CONTACT

info@atomarine.co

WEB

atomarine.co

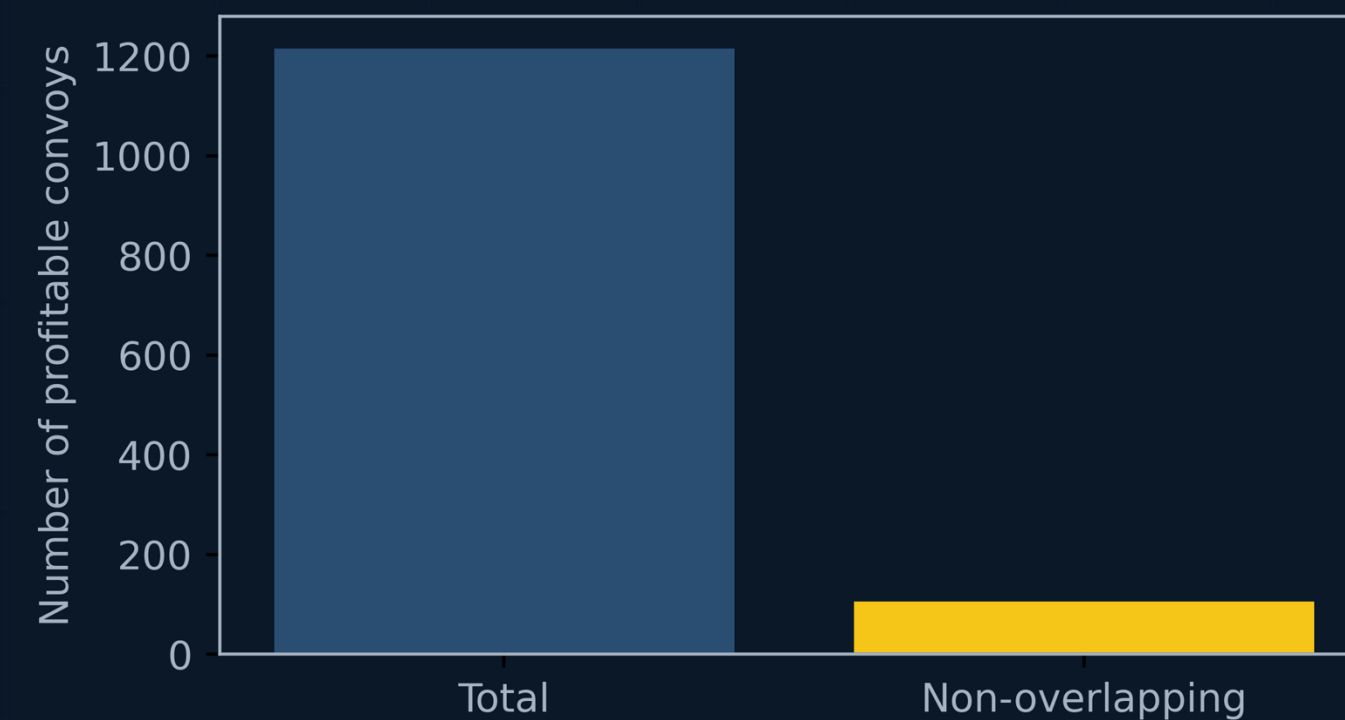
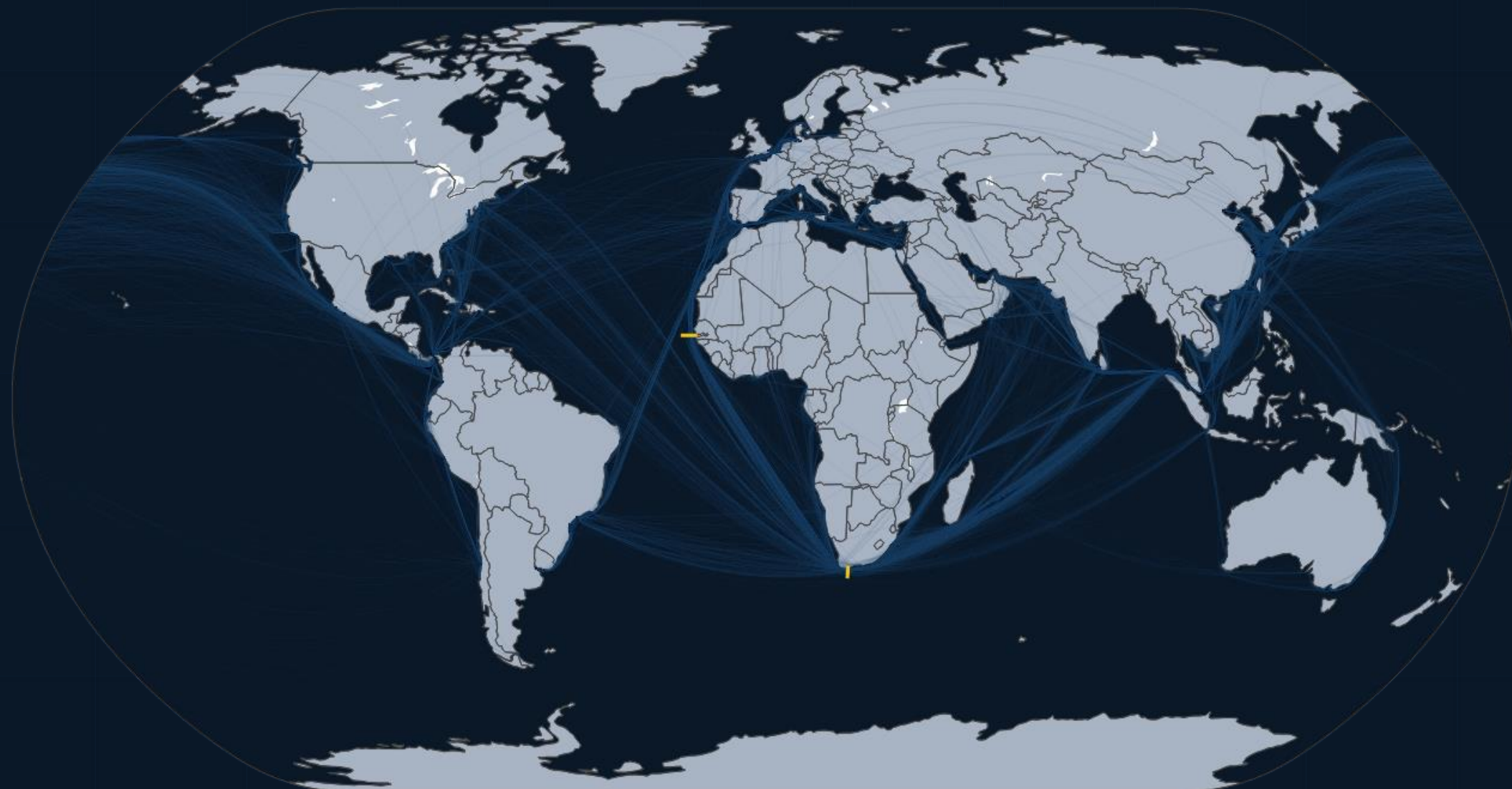
HQ

Cambridge, MA

ROUND

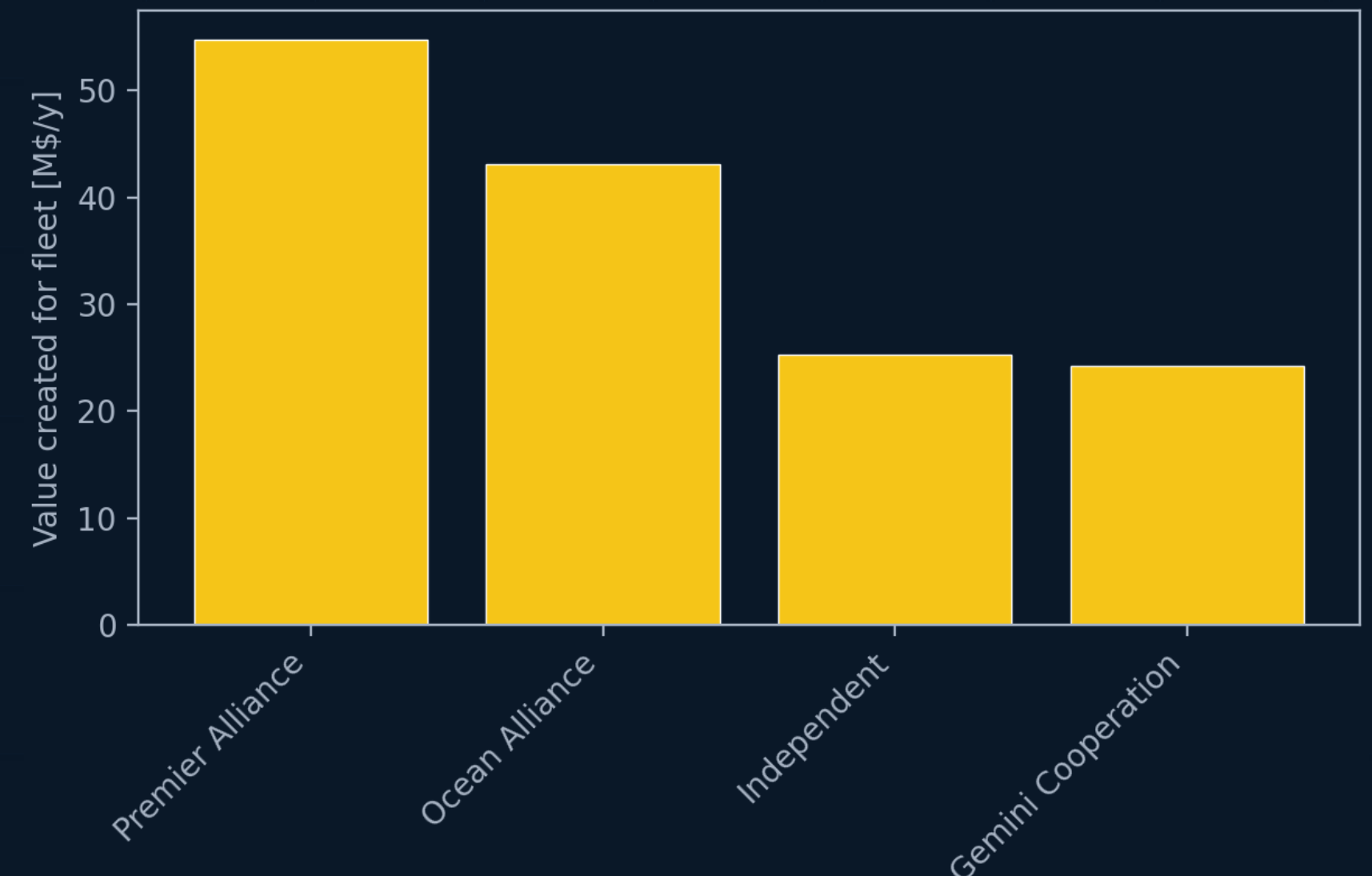
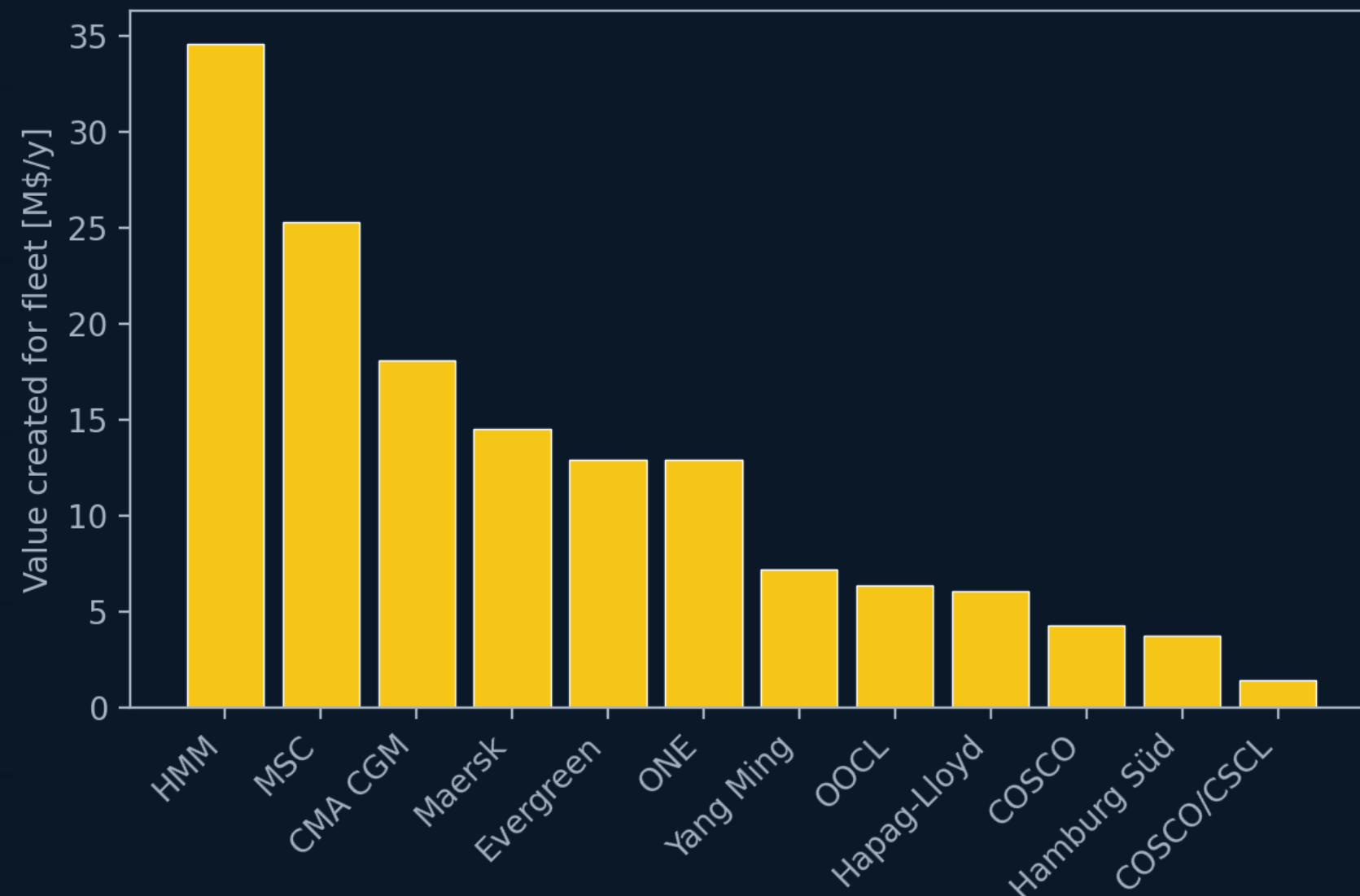
Pre-Seed · \$2–5M

Ship traffic clusters in lanes, making convoy formation feasible



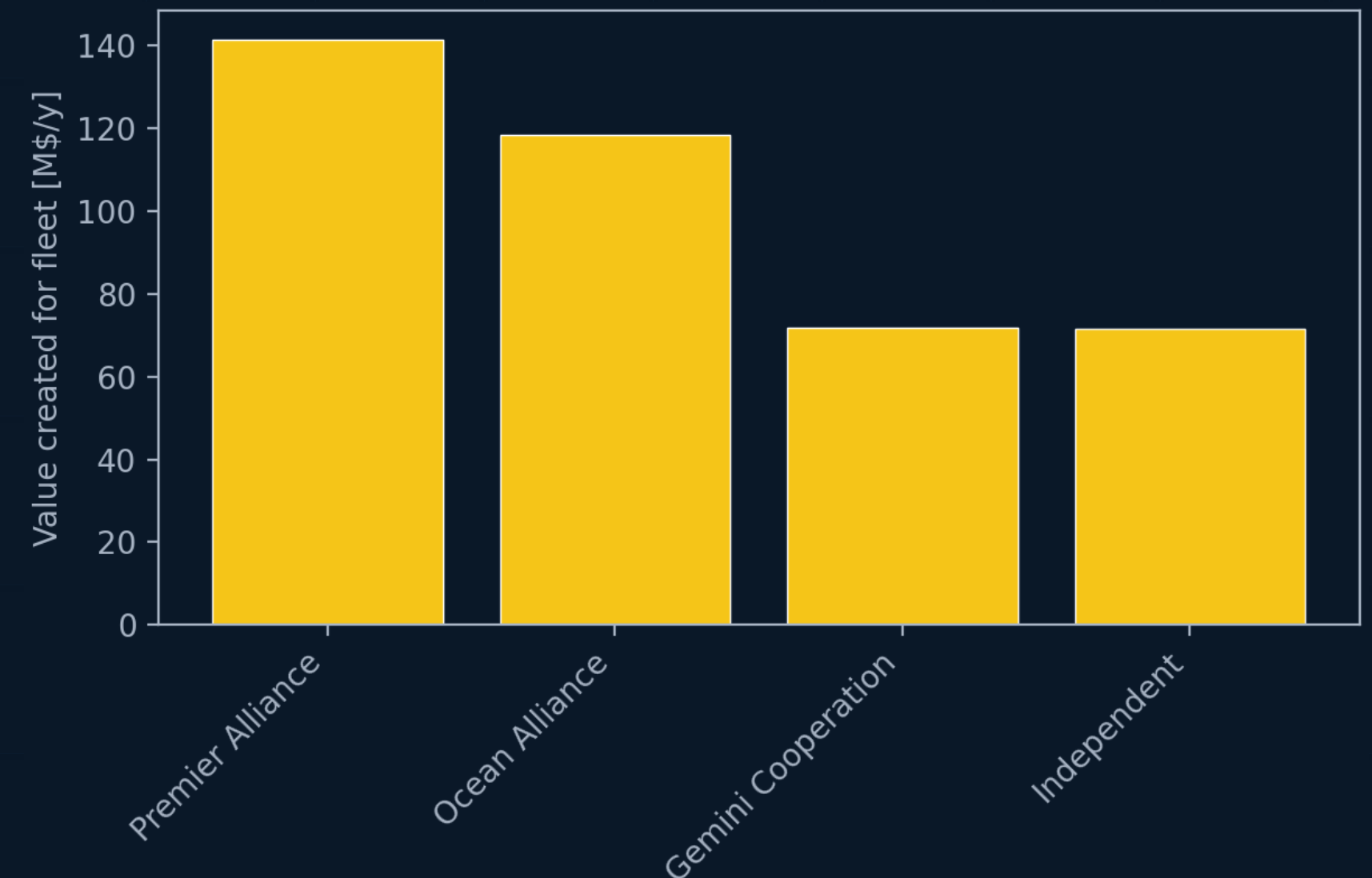
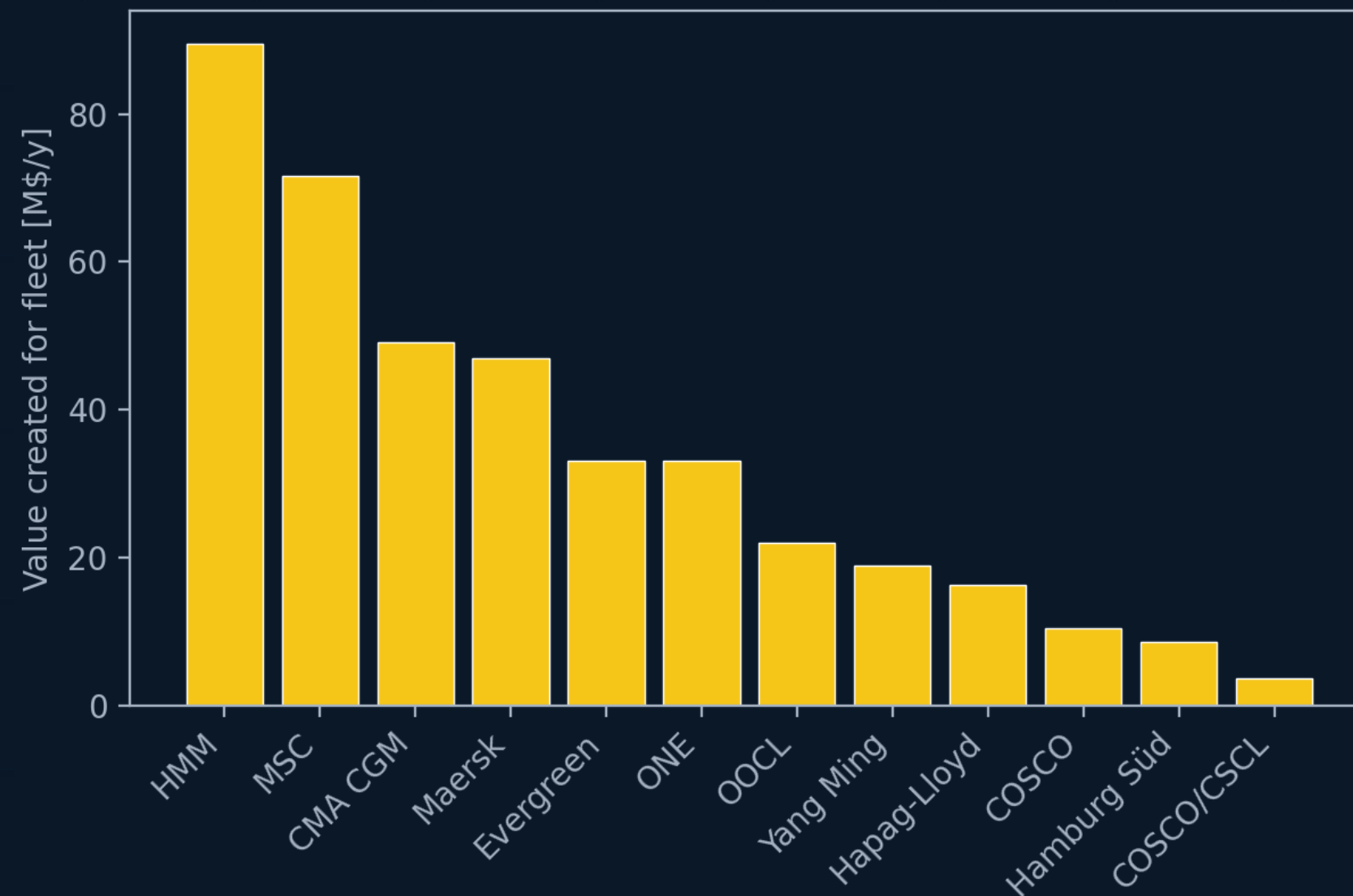
2025 AIS data for >7k TEU ships

Even without carbon tax, we create value for our customers



Value created = offset fuel cost + increased revenue from speedup (+ avoided carbon cost) – LeadShip operational cost - capital repayment for 20 yr LeadShip payback

Carbon taxes represent a large upside



Value created = offset fuel cost + increased revenue from speedup (+ avoided carbon cost) – LeadShip operational cost - capital repayment for 20 yr LeadShip payback

— WHY NOW

The Post-Combustion Era starts now.

Three independent forces. One unmissable alignment.

01

Regulatory certainty

EU ETS and FuelEU make carbon-free shipping inevitable. HFO's volatile pricing and compounding penalties create an untenable cost structure.

02

Technology readiness

SMR licensing accelerating globally. Dynamic cables proven.
Electric-drive ships proliferating. Every component exists today.

03

No alternative path

Every green fuel has failed to scale. Speed mandates are slowing global trade. Nuclear is the only energy source that checks every box.

The window to define the future of shipping is open. It won't stay open.

COMPETITION

The only architecture that bypasses the barriers.

COMPANY	APPROACH	PORT ACCESS	TIMELINE	SCALABLE
Allseas	Onboard SMR	Required	Long	No
Core Power	Floating NPPs	Required	Long	Partial
Maritime Fusion	Compact fusion	Required	Very long	Unproven
SMR Vendors	Near-shore float	Required	Medium	Partial
Atomarine	Offshore NPaaS	Eliminated	Medium	Yes

Atomarine is faster, cheaper, and eliminates all risk for the shipowner.

Four risks. Four mitigations. No PR hedges.

RISK · REGULATORY TIMELINE

Nuclear at sea — under what Flag State, on what schedule?

MITIGATION

Decouple reactor approval from vessel approval. Early Flag-State selection. Technology-taker stance — we ride the SMR vendor's certification.

RISK · TECHNICAL INTEGRATION

At-sea 100 MW HV transfer between moving ships.

MITIGATION

Dynamic HV cables already in service for offshore-wind O&M at similar voltages. Pre-Seed funds independent naval-arch review; Seed funds scaled sea-trial.

RISK · CAPITAL INTENSITY

\$1B+ per LeadShip, 8+ year horizon to cash flow.

MITIGATION

Staged infrastructure capital stack. Take-or-pay capacity contracts back project-finance debt. We don't pre-fund the LeadShip at Seed — we pre-fund commitments.

RISK · PUBLIC PERCEPTION

Nuclear + ocean is a headline waiting to happen.

MITIGATION

Offshore-only operations. International waters. Flag-State jurisdiction. No port calls — ever. Narrative: 'power-sharing vessel,' not 'nuclear ship.'